NATURAL GAS DELIVERY OPTIMIZATION

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ML forecasting + stochastic linear programming to minimize cash-out penalties across daily injection/withdrawal and end-of-month storage constraints.

DATASET
3,410 days

PERIOD
Aug 2015 - Nov 2024

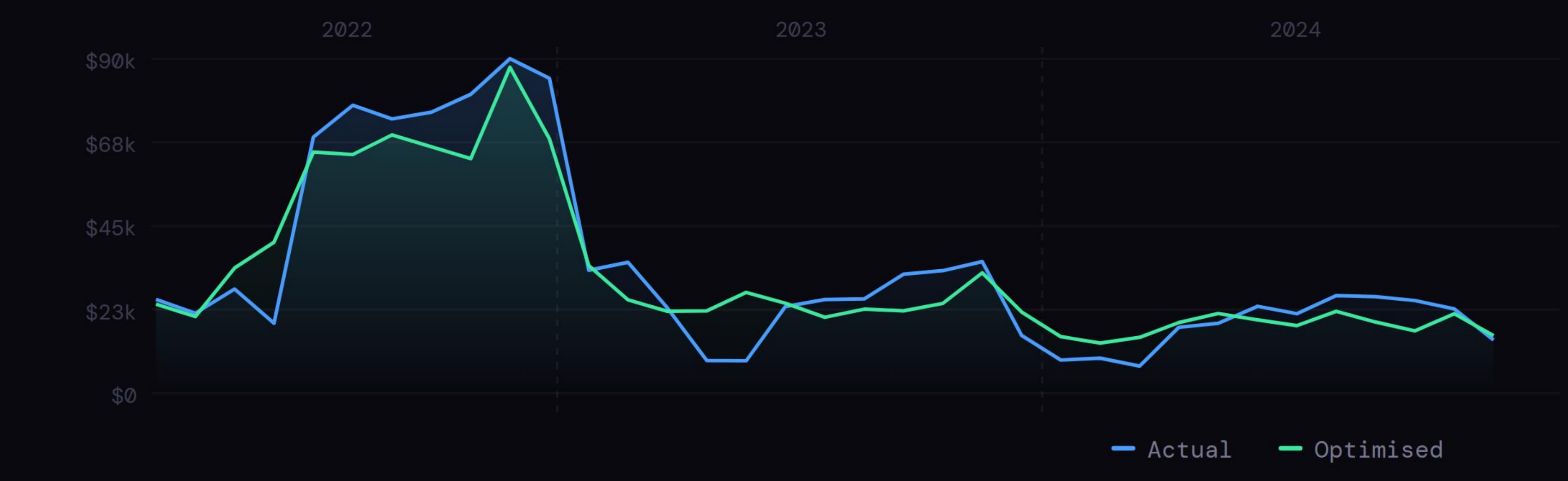
KEY RESULTS

42.3%

Penalty Cost Reduction

ACTUAL PENALTY **$532,491** 420 violation days	OPTIMISED PENALTY **$306,978** 156 violation days
PENALTY SAVED **$225,513** across 3 years	VIOLATION REDUCTION **62.9%** 264 fewer days
TOTAL COST ACTUAL **$11,863,466** base + penalty	TOTAL COST OPT **$11,396,756** $46,671 saved (3.9%)

MONTHLY COST — ACTUAL VS OPTIMISED



2022 — Violations: 128 / 57

2023 — Violations: 156 / 51

2024 — Violations: 136 / 48

CASH-OUT TIERS

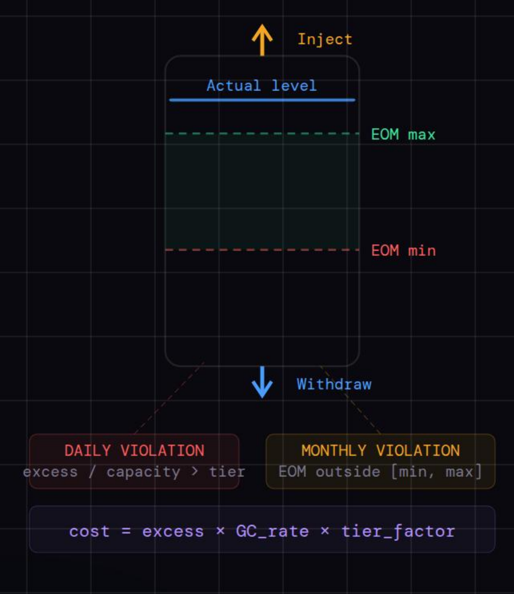
TIER	VARIANCE	PENALTY
T1	0–10%	0%
T2	10–20%	15%
T3	>20%	40%

TIER DISTRIBUTION — ACTUAL: T1 71, T3 302

TIER DISTRIBUTION — OPTIMISED: T1 22, T3 109

TIER 3 DAYS: 302 act, 109 opt

PENALTY EVALUATION



METHODOLOGY

PIPELINE

01 **2-Day-Ahead Blended Forecasting** `GBM+QR`
Blended quantile model (70% GBM + 30% QuantileRegressor) for P10/P50/P95. Features: lags T-3 to T-30, rolling stats, holiday flags, storage %, days-to-EOM. Minimum lag = 2 enforces no leakage.

02 **Quantile Uncertainty Bands** `P10/P95`
Three quantile levels (P10, P50, P95) trained simultaneously. P95 forecast used as conservative usage estimate for LP. Monthly expanding-window retrain across 36 model snapshots (Jan 2022 – Dec 2024).

03 **Adaptive Contract Limit Buffering** `Adaptive`
Month-specific injection buffers (78–85%) and withdrawal buffers (80–88%) shrink effective LP limits. Tighter in summer (inj) and April (wd) where violations historically concentrate.

04 **Cost-Weighted Deterministic LP** `HiGHS`
Minimises Σ c[t]·delivery[t] where c[t] = 1 − α·(inj_norm + wd_norm), α=0.10. Hard flow/storage bounds, soft multi-EOM and mid-month steering constraints. HiGHS solver, 3-level fallback.

05 **Rolling MPC Loop (60-day horizon)** `MPC`
60-day lookahead, act on delivery[0] only. Overshoot guard floors delivery when storage exceeds EOM_max. Hard post-LP clip to raw contract limits. Re-plan every day with updated actuals.

ROLLING LP STRUCTURE (V4)

Forecast Input
```
usage_p95[t] horizon=60d
```

LP Solver (HiGHS)
```
min Σ c[t] · delivery[t]
c[t] = 1 − α·(inj_norm + wd_norm)

var: delivery[t] ≥ 0, t=1..H
solver: HiGHS | α = 0.10
```

Act on delivery[0] only
```
re-plan daily · MPC loop
```

HARD CONSTRAINTS
```
delivery[t] ≤ inj_lim[t]·β_inj + usage[t]
β_inj ∈ [0.78, 0.85] by month

delivery[t] ≥ usage[t] − wd_lim[t]·β_wd
β_wd ∈ [0.80, 0.88] by month

0 ≤ storage[t] ≤ Max Storage
storage[t] = s₀ + Σdelivery − Σusage
```

SOFT CONSTRAINTS
```
EOM_min ≤ storage[eom] ≤ EOM_max
one per month in horizon · buf=6–10%

storage[day15] = (EOM_min+EOM_max)/2
slack = ±6%·capacity mid-month steer
```

POST-LP ADJUSTMENTS

Overshoot Guard: floor delivery if storage > EOM_max

Hard Clip: clip to raw contract inj / wd limits

HORIZON	LAG	QUANTILE	LP MARGIN A
60d	2-day	P95	0.10

FORECAST ACCURACY

89.0%

[P10, P95] interval coverage

P50 FORECAST MAE — 2-DAY AHEAD
229.6 units
vs mean daily usage of 1,736.7 — 13.2% relative error

P10–P95 INTERVAL WIDTH
±515 units
P95 over-forecasts by +538 on average — conservative by design

MODEL DETAILS

Target	Total Usage (direct)
Min lag	T-3 (lag_1 + 2-day shift)
Eff. lags	T-3,4,5,9,16,23,30
Features	23 total incl. holiday flags
Retrain	36 snapshots, monthly expanding

DATA SCIENCE CAPSTONE PROJECT · 2026

Python · scikit-learn · scipy.optimize · HiGHS · Marimo · Altair · Conformal Prediction · Stochastic LP · Rolling MPC